



Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2025-26

Class:	BE	Semester:	VIII
Course Code:		Course Name:	PM

Name of Student:	BARI ANKIT VENOD
Roll No. :	61
Assignment No.:	3
Title of Assignment:	
Date of Submission:	
Date of Correction:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Completeness	5	4
Demonstrated Knowledge	3	3
Legibility	2	2
Total	10	9

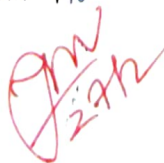
Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Completeness	5	3-4	1-2
Demonstrated Knowledge	3	2	1
Legibility	2	1	0

Checked by

Name of Faculty : Dr. Tatwadashi PN

Signature :

Date :



2. i) A company is developing a college website within 4 months. The project includes requirements gathering, design, content development, coding, testing and deployment. The team consists of a project manager, designer, developer, content writer and tester. question - As the project manager, illustrate how would you prepare a work breakdown structure and a linear responsibility chart for this project, explain using the given scenario.

→ A work breakdown structure is a hierarchical decomposition of the total project work into smaller, manageable components called work packages. for the college website development project (4 months), the WBS would be structured as follows -

1.1 Requirements gathering

- stakeholder meetings
- requirement documentation
- requirement approval

1.2 Design phase

- UI/UX design
- wireframes & mockups
- design approval

1.3 Content development

- content planning
- content writing
- content review & editing

1.4 website development (coding)

- front-end development
- back-end development
- database set-up
- Integration

1.5 Testing

- Bug fixing

- How WBS helps in this scenario -
 - clearly defines project scope
 - breaks work into manageable tasks
 - helps in scheduling and cost estimation
 - avoids scope creep
 - classifies deliverables for each team member.
- Linear responsibility chart -
 - A linear responsibility chart assigns role to each task -
- Team members -
 - PM - project manager
 - DES - designers
 - DEV - developers
 - CW - content writer
 - TEST - Tester
- we use common responsibility codes -
 - R = Responsibility
 - A = Accountability
 - C = Consulted
 - I = Informed
- Explanation of roles -
 - ① Project manager (PM)
 - Accountable for requirements
 - Approves final output
 - ② designer - responsible for UI/UX
 - supports deployments
 - ③ content writer - Responsible for all website content
 - ④ Tester - Responsible for quality assurance.

2) A project consist of the following activities with three time estimates, optimistic (t_o), most likely (t_m), pessimistic (t_p).

construct the network diagram for the given project using the above activity data, determine the critical path and calculate the expected project completion time.

→ ∴ we use PERT expected time formula -

S1 - Calculate expected time -

$$t_e = (t_o + 4t_m + t_p) / 6$$

Activity	t_o	t_m	t_p	t_e
1-2	2	6	9	$(2 + 24 + 9) / 6 = 6$
1-6	2	5	8	$(2 + 20 + 8) / 6 = 5$
2-3	6	12	18	$(6 + 48 + 18) / 6 = 12$
2-4	4	5	6	$(4 + 20 + 6) / 6 = 5$
3-5	8	11	14	$(8 + 44 + 14) / 6 = 11$
4-5	3	7	11	$(3 + 28 + 11) / 6 = 7$
6-7	3	9	15	$(3 + 36 + 15) / 6 = 9$
5-8	2	4	6	$(2 + 16 + 6) / 6 = 4$
7-8	8	16	18	$(8 + 64 + 18) / 6 = 15$

S2 - Identified paths -

① 1-2-3-5-8 = $6 + 12 + 11 + 4 = 33$

② 1-2-4-5-8 = $6 + 5 + 7 + 4 = 22$

③ 1-6-7-8 = $5 + 9 + 15 = 29$

S3 - Critical path -

∴ longest path = 33

∴ critical path = 1-2-3-5-8

∴ Units = 33 times units

Q. 27 Construct network diagram and estimate critical path.

→ S1. Network structure -

start activities - A (6), B (5)

Then, C (after A) = 4

D (after A) = 3

E (after A) = 2

F (after E) = 4

G (after D & F) = 14

H (after B & C) = 12

I (after G & H) = 3

S2. forward pass -

$$A = 6$$

$$B = 5$$

$$C = 6 + 4 = 10$$

$$D = 6 + 3 = 9$$

$$E = 6 + 2 = 8$$

$$F = 8 + 4 = 12$$

$$G = \max(9, 12) + 14 = 12 + 14 = 26$$

$$H = \max(5, 10) + 12 = 10 + 12 = 22$$

$$I = \max(26, 22) + 3 = 26 + 3 = 29$$

∴ Project duration = 29 months

S3. Identifying critical path -

∴ Trace longest path -

$$A \rightarrow E \rightarrow F \rightarrow G \rightarrow I$$

$$\therefore 6 + 2 + 4 + 14 + 3$$

$$= \underline{\underline{29}}$$